A Fuzzy-Logic-Based Student Learning Assessment System for Outcome-Based Education

Abdul Aziz, Md. Asaf-uddowla Golap, and M. M. A. Hashem

Abstract*—*Contribution: **This research designs a student evaluation framework integrating the fuzzy-logic system that assesses the student's performances in the soft boundary system for outcome-based education (OBE), measuring the course learning outcome (CLO) and program learning outcome (PLO). The framework fills the gap between conventional grading methods and offers insights into learning for course assessment and continuous development.**

Background: **A well-established evaluation technique is a requirement to deliver a productive, skilled, worthy, and compatible student and faculty. Moreover, OBE, with a documented and structured academic curriculum, has to ensure the accreditation of an academic program.**

Research Questions: **What are the drawbacks of traditional student evaluation techniques? Does the proposed system work as a better, more reliable, and meaningful student evaluation method?**

Methodology: **To assess, it considers the final examination paper containing several questions and continuous assessment comprising a few items like class tests, quizzes, viva voce, homework, etc., where the course teachers and moderators assign marks on these questions and items considering the CLOs, learning methods, and Bloom's taxonomy. Then, the framework records and tracks the ratio of earned marks to assigned marks for the fuzzification, while the defuzzification computes the values indicating the CLOs and PLOs earned by a student.**

Findings: **The results study cases for 40 courses of a particular student and analyze statistics for 100 students from the consecutive eight semesters. This fuzzy-logic-based evaluation technique is fairer, reliable, and unbiased to the learners and greatly helps to get accreditation and recognition for the degree worldwide.**

***Index Terms*—Course learning outcome (CLO), fuzzy evaluation system, outcome-based education (OBE), program learning outcome (PLO), student performance assessment.**

I. Introduction

IN MODERN days, quality assurance is one of the most vital matters for students in their academic institutions, where a decent performance evaluation technique can play an important role. Bangladeshi universities have different evaluation techniques which are not very reliable. On the other hand, the degrees offered by most of the departments of those institutions are not accredited all over the world. The graduates from these nonaccredited programs of the universities face several problems outside the country for their higher studies or career purposes. Due to these problems, universities are trying to maintain their academic accreditation. Accreditation is the official recognition and acceptance of an educational institution or a program, which is assessed against established criteria and standards. Accreditation helps students, teachers, and graduates in different ways. The graduate from an accredited institution has an opportunity to be promoted in license, employment, global mobility, and easily apply, register, and certify for school. Normally, the universities have two types of academic accreditation processes [1]. First, institutional accreditation means that it accredits all of the programs of the institutions. Second, specialized accreditation accredits certain programs of that institution. This research falls under the second category of accreditation. At Khulna University of Engineering and Technology (KUET), the Department of Computer Science and Engineering (CSE) is trying to accredit its program. The ABET is a nongovernmental, nonprofit organization that works for accredited academic programs. ABET accredits four types of programs, such as applied and natural science, engineering, engineering technology, and computing programs [2]. They have some basic criteria and a set of standards to follow throughout the whole accreditation process. Among those, a fair evaluation of students' academic performances is one of the most important criteria. The University Grants Commission (UGC) of Bangladesh is working on accrediting its universities. Following that, the department of CSE in KUET is much more in progress in its program accreditation procedure.

A. What Is OBE?

Outcome-based education (OBE) is a prerequisite to accrediting an academic program. OBE means the education opening with a complete structure of program indicating what is essential for the students to do and then designing the curriculum and an evaluation technique so that OBE can be determined whether it is happening or not [3]. For each of the academic programs ensuring OBE, ABET has a set of criteria that have to be learned by students after passing the program, known as a program learning outcome (PLO). This PLO should have been documented and acknowledged by both students and teachers, which will help them to acquire

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Abdul Aziz and M. M. A. Hashem are with the Computer Science and Engineering Department, Khulna University of Engineering and Technology, Khulna 9203, Bangladesh (e-mail: abdulaziz@cse.kuet.ac.bd; hashem@cse.kuet.ac.bd).

Md. Asaf-uddowla Golap is with the Institute of Information and Communication Technology, Khulna University of Engineering and Technology, Khulna 9203, Bangladesh (e-mail: asaf.golap@iict.kuet.ac.bd).

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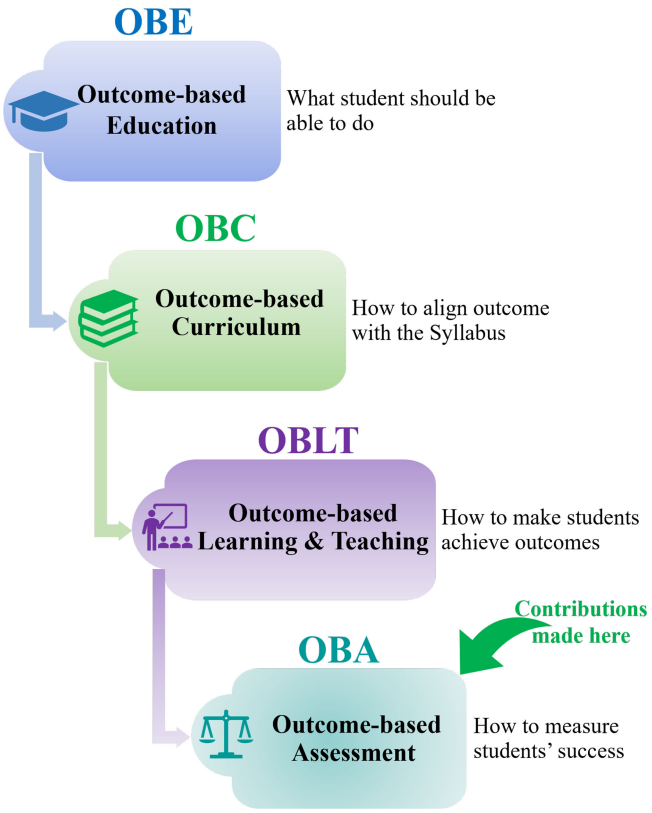


Fig. 1. Components of OBE: (1) what students should be able to do, (2) how to align outcomes with the syllabus, (3) how to make students achieve outcomes, and (4) how to measure students' success.

program educational objectives easily. The PLO means the skills, abilities, and attributes that the learner will be capable of doing upon graduating from that program [4]. The CLOs are the statements that a student will be able to learn after the end of the course.

B. Outcome-Based Assessment

OBE is a broad domain in the research field of educational development. It refers to an educational approach that prioritizes a clear knowledge of what students should learn and achieve. This includes identifying the skills and knowledge they should possess upon completing their graduation. For OBE implementation, three more components have to be also established illustrated in Fig. 1. Outcome-based curriculum (OBC) development ensures the alignment of outcomes with the syllabus. OBC explains what students should be able to achieve by the end of a course. Outcome-based teaching and learning (OBLT) refers to a student-centered education approach focused on students, wherein the desired learning outcomes of the program are explicitly defined for students to attain. Based on the syllabus, OBLT ensures a teaching and learning environment so that students can achieve outcomes. Finally, outcome-based assessment (OBA) is required to evaluate the students' outcomes (SOs). OBA is an evaluation approach to assess knowledge, skills, or competencies based on specific predefined outcomes or objectives. In this article, the authors proposed an OBA system to measure the SOs.

The proposed fuzzy inference system determines the student's level of attainment, making the evaluation outcome-based. This gains the OBE, measuring the CLO and PLO of each of the courses associated with the evaluation procedure. CLOs are those statements indicating skill, knowledge, etc. that will be learned after the completion of the course taught by the course teacher, and PLOs are those statements that are used to clarify the expectations from the students to know and what they can attain or perform during the time of graduation in terms of cognitive, psychomotor, and affective domains [5].

C. Bloom's Learning Levels

Learning is the continuous process of growing and changing in the brain's architecture, resulting from the various ways of taking in information and then processing, connecting, cataloging, using, and sometimes getting rid of it. The creature learns from everywhere. People are learning mental skills, developing their attitudes, and acquiring new physical skills by performing daily living activities. The learning method is generally categorized as cognitive, affective, and psychomotor in three domains. Each of the domains has multiple levels of learning, which process from a surface or lower level of learning to a deeper or more complex level of learning [6]. The cognitive includes learning skills mainly associated with the thinking or memorization process. There are six levels of its complexity: 1) remembering; 2) understanding; 3) applying; 4) analyzing; 5) evaluating; and 6) creating, as proposed by Anderson and Krathwohl [52]. The affective domain encompasses the emotional responses to things like feelings, values, admiration, enthusiasm, motivations, and attitudes. It is classified into five levels, which include receiving, responding, valuing, organizing, and characterization. Psychomotor goals are those that are particular to specific physical functions, reflex activities, or interpretive motions. Simpson [53] proposed seven levels of this domain that fall under this heading: 1) perception; 2) set; 3) guided response; 4) mechanism; 5) complex overt response; 6) adaptation; and 7) origination. These different levels of each domain have their own action verbs [7] that are used to design the course CLOs and set question papers accordingly. Moreover, the instructions from [8] are also followed for making the CLOs well constructed for each course.

D. Limitations of Existing Evaluation Techniques

In a traditional learning environment, the instructional focus is primarily centered around the teacher, emphasizing their role as the primary authority figure. Conversely, in an OBE learning environment, the teacher assumes the role of a partner or facilitator, fostering collaborative learning experiences. While traditional methods prioritize the input provided by teachers, OBE-based methods rely on the output generated by learners, emphasizing active participation and engagement. Traditional approaches to education tend to be rigid and

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instructor-centered, exerting control over the learning process. In contrast, OBE-based methodologies are characterized by flexibility and empowerment, allowing learners to take ownership of their learning journey. OBE-based education places emphasis on progress and holistic learning outcomes, whereas traditional methods often prioritize the attainment of specific products or outcomes. Assessment in an OBE context typically employs criterion-referenced measures, aligning with the objectives and standards set forth in the curriculum. Conversely, traditional assessment methods often utilize norm-referenced measures, comparing individual performance to that of a predefined norm or standard. The overarching aim of OBE is to cultivate and enhance learners' abilities and skills, focusing on competency development. In contrast, traditional methods tend to be content-centered, focusing primarily on the delivery of subject matter content.

In most Bangladeshi institutions, student performance evaluations are done on the percentage of getting marks. For example, if mark $\geq$ 80% then grade is A^+, 75% $\leq$ mark $\leq$ 79% then grade is A, 70% $\leq$ mark $\leq$ 74% then grade is A^-, 65% $\leq$ mark $\leq$ 69% then grade is B^+, and so on [9]. This grading system, using the absolute mark ranges, is sometimes fair and sometimes unfair. Because the evaluation of two students getting marks of 80 and 99 is the same in grade. Surprisingly, 79 and 80 are not the same though the difference is smaller than the difference between 80 and 99. That is why there need for a fuzzy-based evaluation other than classical methods. In the classical system, there are only two possible outcomes (0 and 1) exist, where 0 indicates the absolute false value and 1 indicates the absolute truth value. But in the fuzzy system, multiple possibilities are present between 0 and 1, which are partially true and partially false. Ewing [10] showed the impact of relative expected grades on students of Washington University (WU) using the data provided by the Office of Educational Assessment (OEA). Relative grading means x% of students should be assigned to the highest grade, y% assigned to the second highest grade, and so on. Similarly, z% of students are assigned to fail. In this grading system, this z% of students may have the ability to pass the examinations. Due to the relative grading, they cannot, which is unfair and may be monotonous to students. Hussain et al. [11] showed an impact evaluation on engineering students. They assessed only ABET SOs to visualize the impact using the regression analyses and did not provide course outcome (CO) evaluations. Moreover, this range of grading and relational grading does not provide a precise level of gaining, reliability, and robustness other than biases.

The primary objective of this article is to address the limitations inherent in existing evaluation methodologies by proposing a comprehensive learning assessment framework. So, this article presents a structured learning assessment framework adapted specifically for OBE. Specifically, the proposed approach measures the attainment of specified learning outcomes by structuring examination papers in alignment with course learning outcomes (CLOs) and PLOs, instead of just measuring academic achievement and grades. Moreover, the framework uses a fuzzy-based grading system as opposed to conventional absolute grading methodologies to increase fairness and reliability.

E. Major Contributions

The proposed evaluation technique provides a fair level of attainment for both students and teachers. The summarized major contributions to this article are as follows.

1) Designing an OBE-based student evaluation model based on the fuzzy-logic system, which assesses the student's performance in the soft boundary system.
2) Assessing the proposed evaluation model experimentally and theoretically.
3) Introducing a new intelligent student assessment technique, generating a more meaningful grade for students than a hard boundary system.

II. Literature Review

Recently, researchers have emphasized the development of student performance evaluation techniques.

A. Non-Outcome-Based Evaluation Methods

Chen et al. [14] showed the impact of artificial intelligence (AI) on education. They have restricted their study to the application and effect in administration, instruction, and learning. In the study, they have discussed the scenarios of AI education as assessment of students, grading or evaluation of exam papers, personalized intelligent teaching, online or mobile remote education using adaptive learning methods, image recognition, computer vision, data mining, or Bayesian knowledge inference, edge computing, etc. There were some adverse impacts of AI on education declining academic integrity and defrauding using paper churning and paper mill serving aided by AI. Gupta [25] gave a demonstration of the influence of Information and Communication Technologies (ICTs) in conducting software engineering (SE) courses with project-based learning (PBL) and design-based learning (DBL). However, he showed the demonstration for only an undergraduate course for two consecutive years, rather than providing a complete view of how DBL and PBL are in every part of the faculty. Saha [26] proposed a simulation-based practical classroom with a developed educational process for the department of electrical engineering students for managing problems of large class sizes in front of the expansion of physical laboratories, and mentioned that it can be as effective as a traditional approach accommodating and practical enormous class size efficiently. They used industry-standard software to teach students with an established assessment method and a feedback model for evaluation.

Deo et al. [17] set an AI-based model for undergraduate student performance prediction only for engineering mathematics courses. They used random forest and extreme learning machines with a second-order Volterra model based on mathematics to observe the feasible impact of continuous assessment on the weighted score, and grades in mathematics courses for engineering students of several years. Mitchell et al. [27] introduced a problem-based learning (PBL) approach, which

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is enormously useful in inspiring the improvement of the transferable efficiency that is valued by staff, observed students becoming adept at group working and developing their communication skills by keeping them engaged in the discussion with PBL. Wang and Chen [28] proposed a fuzzy-logic-based method for evaluating students' answer scripts. Satisfaction level associated with the degrees of confidence provided by the evaluators represented by a triangular fuzzy membership function. Then, different arithmetic operations between fuzzy numbers were used to evaluate the total mark of each student. Finally, they calculated the confidence level of the total mark of each student. They also used the optimism index to represent the degree of optimism of the answer script evaluator.

B. Outcome-Based Evaluation Methods

Shafi et al. [12] proposed an ABET-defined student OBA through a hypothetical technique for computer science and computer information systems programs. They presented a general evaluation framework for ABET accreditation. Syeed et al. [13] designed a concrete and comprehensive academic framework for OBE implementation at the department of CSE, Independent University of Bangladesh. They suggested class tests (CTs), lab works, lab tests, progressive projects, and project reports as continuous assessment parameters, and midterm and final terms as summative assessment parameters. Overall, they discussed CLOs, PLOs, and other OBE-related terms, and finally proposed a guideline on how OBE affects students' assessment. Shanableh [21] demonstrated CO and program outcome (PO)-based evaluation systems for IT-facilitated students. They did this for the Civil Engineering Program, which allows a level of achievement based on CO and PO. Chen et al. [29] determined the online-based short CO by behavior-based machine learning feature selection collected from the students' learning and interaction with one another in a social learning network. Chandna [18] proposed a method for evaluating the performance of an individual weak student. He measured the COs and POs on different levels, such as high (H), medium (M), and low (L), and set a goal of attainment so that the failed students could address their shortcomings to improve themselves further. Aziz and Hashem [30] introduced an outcome-based student performance evaluation method using the fuzzy inference system. They considered only five courses from a random semester for one student for the result generation, which is not enough to understand whether an evaluation system is better or not. Unlike this one, the proposed system considers not only 40 courses of a student but also generates the statistical changes of an additional hundred students. Spady [31], the father of OBE, discussed the OBE in detail in his book published in 1994. He picked up several critical issues and answers to those issues in his book, including the major trends of OBE and how the OBE implementation affects institutes, students, etc. Faisal et al. [24] introduced a systematic method to calculate the COs and SOs of any program. They considered the mapping between the subject topics and COs, COs, and SOs. But they just showed an example of their model, other than running it on the dataset.

Dawood et al. [32] implemented an assessment plan for the computer science program of King Faisal University, Saudi Arabia. They made a rubric cube for the assessment mechanism for ABET standard SOs. Abou-Zeid and Taha [33] proposed the requirements for the program accreditation process. They emphasized more on the challenges have to face on this journey, such as faculty members are not enthusiastic about data collection, analyses, form preparation, inadequate training management, heavy workload, understaffing, and a deficiency of appropriate collaboration from the administration. Calderón et al. [34] showed a successful accreditation process for the electrical engineering program for two separate campuses of Caribbean University and discussed the deficiency of literature to design the accreditation strategies. Zaini et al. [35] designed an online OBE framework containing a central data repository for an institution, allowing online concurrent transactions among multiple users. Mahmood et al. [36] described the faculty assessment report experience to assess and evaluate student performance for B.Sc. in the electrical engineering program at Iqra National University, Peshawar of Pakistan. Prasad and Reddy [37] proposed a computer-based active teaching methodology (CATM). They designed CATM and measured COs and POs for several courses, such as Micro Processors and Micro Controllers, and Embedded Systems, and Linear Digital Integrated Circuit Applications courses for engineering students. Lakshmi [38] introduced a course and PO-based teaching methodology for two different microwave and radar courses. They mapped between the COs and POs but gave more importance to the COs.

Akir et al. [39] compared the OBE and the conventional grade-point average teaching-learning approach and found that OBE is better for student evaluation. Barney et al. [40] developed a rubric-based evaluation technique for self-assessment and oral feedback of teachers which improves students' learning outcomes. The students and teachers both have to invest more time in their skill improvement. Rasha and Shatakumari [22] reviewed a trend and discussed the origin, nature, advantages, disadvantages, pitfalls, and guidelines to implement OBE. Bouslama et al. [41] designed an academic model on OBE and showed its effect on students' assessment, student learning, and curriculum development. Hassan [23] noted down the challenges that have to be faced to implement OBE in Bangladeshi universities as OBE is not practiced in almost all universities. Saad and Haque [42] proposed an OBE-based evaluation measuring the COs of the courses for the Department of Electrical and Electronic Engineering (EEE). They measured the COs in three different ways by absolute score, normalized score, and weighted normalized score, and mapped them with the appropriate POs. However, they did not provide any methodology to attain the POs.

C. Fuzzy-Based Evaluation Methods

Hameed and Sørensen [15] and Saleh and Kim [16] presented a fuzzy-based evaluation system. They fuzzified their system three times to generate the grade and finally compared their evaluation result with the traditional evaluation.

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TABLE I
SUMMARY OF THE RELATED WORKS

Authors	Methods	Dataset	Weakness/Remarks
Shafi [12] *et al.*	holistic approach	self-made	not much information is available for implementation mechanics, and this results in confusion and wastage of resources. Also, scarcity of available literature outlining the methodology and implementation of successful accreditation approaches.
Syeed [13] *et al.*	deterministic approach	no dataset	just proposed a theoretical framework for OBE implementation without showing enough case studies.
Chen *et al.* [14]	adaptive learning method, image recognition, computer vision, Bayesian knowledge inference, edge computing	self-made	declining of academic integrity and defrauding using paper churning and paper mill serving aided by AI.
A. Aziz *et al.* [9]	fuzzy logic system	self-made	outcome-based evaluation was not considered.
Hussain *et al.* [11]	Regression analyses	SO data	only evaluate determining the student's outcome.
Hameed [15], Saleh [16] *et al.*	Fuzzy logic system	self-made	provides a system with less degree of freedom.
Deo *et al.* [17]	random forest, extreme learning machine (ELM)	university benchmark data	student performance in an advanced engineering mathematics course, *ELM* registered slightly larger errors.
Chandna [18]	deterministic approach	self-made	does not provide a details explanation of how weak students develop themselves.
Ewing [10]	ordinary least-squares (OLS) model	WUs OEA data	students having the ability to pass the test due to the relative grading, they can't, and this is unfair.
Ma *et al.* [19]	fuzzy set approach	self-made	students' learning outcome is not discussed, rather, assessment criteria and setting the weights are student-centered.
Varghese *et al.* [20]	fuzzy logic system	self-made	all the cases of fuzzy rules are not used to implement or calculate the CLOs.
Shanableh [21]	deterministic approach	self-made	case studied with only two courses generating their outcomes, which is not enough for making a decision.
Rasha *et al.* [22]	holistic approach	no dataset	just reviewed the trend of outcome-based education, discussing the advantages, disadvantages, pitfalls, and guidelines of implementing it.
Hassan [23]	holistic approach	no dataset	discuss the overall scenario of OBE in Bangladeshi universities without showing any sample assessment.
Faisal *et al.* [24]	deterministic approach	no dataset	no case studied, but shows an example only.
Gupta [25]	deterministic approach	no dataset	demonstrate for only one course and consider an online exam, which is tough to maintain the quality of students' evaluation.
Saha [26]	not mentioned	no dataset	just introduce an industry-standard software to educate students without showing any case study of evaluation.

Varghese et al. [20] introduced OBA using the fuzzy-logic system. They fuzzified each of the COs on five input levels and generated an output CO on five levels. In all the possible cases, fuzzy rules were not implemented in their system. Aziz et al. [9] introduced a fuzzy-based grading system considering the final examination papers assessment, including part-A and part-B, and continuous assessment, including the total no. of classes, time spent on class, and CT. This evaluation was not followed by an OBA. However, in addition to the amount of time spent in class and the total number of attendances (ATTs), the suggested evaluation system includes several other crucial assessment indicators. These extra criteria provide a more comprehensive and holistic evaluation of student performance, including parameters, such as viva voce (VV), quiz, homework (HW), spot test (ST), etc. Mreła et al. [43] described a method for learning OBA using fuzzy relations. Two types of applications were considered there. Type-1 sets a cut score for all learning outcomes and must attain the score to pass the examinations. Type 2 did not mandate experts to comply with a single level of outcome score. Ma and Zhou [19] represented a fuzzy set-based approach to evaluate students' performance, which turns the learning process from teacher-centered to student-centered. Students' performance grades are evaluated in fuzzy grades in place of the normal grade point evaluation. To facilitate comprehension, Table I provides a comparative summary of the pivotal existing works. In sum, the decision to prioritize this research reflects a strategic response to the limitations of existing evaluation methodologies, offering a promising avenue for advancing the fairness and alignment of student performance assessment with contemporary educational goals.

III. MATHEMATICAL MODEL AND METHOD

A. System Architecture

To evaluate the student's performance, considering CLOs, the system implements OBE. The methodology is described for Bangladeshi Universities. Especially figures it out for the Department of CSE, KUET. In most Bangladeshi universities, the evaluations are done based on the mid-term exam, final exam, continuous assessment, HW, assignment, ST, CT, VV, presentation, quiz, project work, presentation skills, lab test, etc. The system architecture of the proposed evaluation technique is shown in Fig. 2, and a learning algorithm is also developed and summarized in Algorithm 1.

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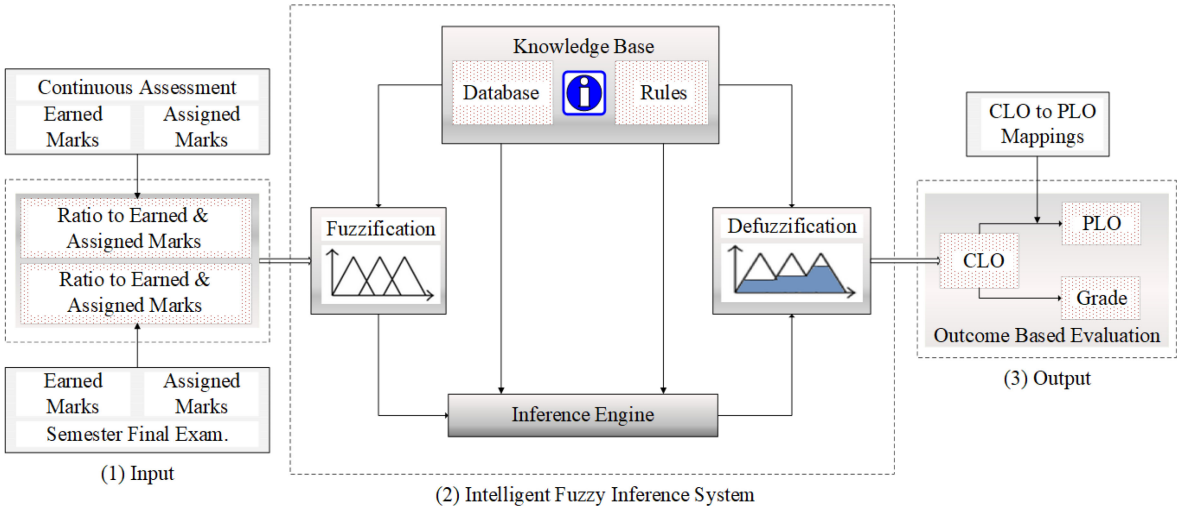


Fig. 2. Proposed evaluation system for OBE. (1) Input is the ratio to the earned marks and assigned marks based on course CLOs, (2) intelligent FIS for fuzzification of the ratio using the inference rules, and (3) output is CLO attainments of a course from defuzzification, the CLOs and mapping values return the PLO attainment, and grade of the course is generated from the CLO attainments.

The system has two parts, considering the semester final examination, which includes several questions, and continuous assessment, which includes a few items like CTs, VV, quizzes, STs, HW, and ATT to assess. Initially, the marks are assigned to each of these questions and items according to the CLOs satisfying Bloom's taxonomy [44], the learning methods [7], and the ABET standards by the course teachers and question moderators panel of the university. Then, the earned marks are recorded and tracked by the course teachers on corresponding questions and items based on CLOs. The ratios between the earned and the assigned marks are then fuzzified. Finally, the defuzzification returns the crisp values indicating the CLOs of the corresponding courses. Using those CLOs, the PLOs of the program are demonstrated and generate the corresponding grades.

B. Data Preparation

For each of the courses, the dataset has a similar structure, having two parts. The first part is for the semester final examination, and the second part is for the continuous assessment. The semester final part contains q (total no. of questions considered for the evaluation in the semester final) number of columns, and the continuous assessment part contains p (total no. of items considered for the evaluation) number of columns. Both of the parts contain n (total no. of CLOs of that course) number of rows. From the real-life teaching and evaluation experiences, the authors have prepared the dataset for this research. Each of the data files contains the assigned marks for each of the questions and items, as well as the earned marks on corresponding questions and items in the semester final examination and continuous assessment. The dataset also includes the mapping values between the n CLOs and $m = 12$ PLOs for each of the courses. The mapping value ranges from 1 to 5, which means the lower the value, the lower the mapping weight, and the higher the value, the higher the mapping weight between the corresponding CLOs to PLOs. The mapping value zero (0) means no relation between the CLOs and PLOs. The dataset was initially created for one student, but later, for analyzing the statistics, it was extended for an additional 100 students.

C. Mathematical Background and Model for CLO Evaluation

The system considers two evaluation parameters. First, the semester final examination consists of several questions and their answers. This is called the summative assessment parameter. Second, the continuous assessment consists of the CT, VV, quiz, HW, ST, ATT, etc. These are called the formative assessment parameters. For laboratories, assessment can be done using VV, quiz, HW, lab test, report, project showcasing, assignment, presentation, etc. In addition, more performance assessment parameters (rubrics) can be added as a generalized system has been proposed. The semester final contains q no. of questions of total marks k which have to be answered by the students, represented by matrix Q as (1) of dimension $1 \times q$

$$Q = [q_i] = [q_1 \quad q_2 \quad q_3 \quad \cdots \quad q_q], 1 \times q \tag{1}$$

where q_i is the total marks assigned on question i based on the CLOs such that

$$\sum_{i=1}^{q} q_i = k. \tag{2}$$

Similarly, the continuous assessment consists of p no. of items of total marks h and marks assignment is represented by matrix P as (3) of dimension $1 \times p$

$$P = [p_i] = [p_1 \quad p_2 \quad p_3 \quad \cdots \quad p_p], 1 \times p \tag{3}$$

where p_i is the total marks assigned on item i based on the CLOs such that

$$\sum_{i=1}^{p} p_i = h. \tag{4}$$

From the system architecture shown in Fig. 2, the input system consists of two parts. The first one is the earned marks

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Algorithm 1: Outcome-Based Student Assessment System Using Fuzzy Logic

Input: n-number of CLOs, q-number of questions in semester final exam, and p-number of items in continuous assessment, m-number of PLOs, earned marks, and assigned marks, CLO–PLO mappings

$Q = [q_1, q_2, q_3,..., q_q]$ /* Final exam question matrix, dimension $1 \times q$ */
$P = [p_1, p_2, p_3,..., p_p]$ /* Continuous assessment matrix, dimension $1 \times p$ */
$S = [s_{ij}]$ /* Assigned marks to each question based on CLO, dimension $n \times q$ */
/* where, s_{ij} denotes the marks assigned to CLO_i on question j */
$C = [c_{ij}]$ /* Assigned marks to each item based on CLO, dimension $n \times p$ */
/* where, c_{ij} denotes the marks assigned to CLO_i on item j */
$A = [a_{ij}]$ /* a_{ij} denotes the marks earned to CLO_i on question j, dimension $n \times q$ */
$B = [b_{ij}]$ /* b_{ij} denotes the marks earned to CLO_i on item j, dimension $n \times p$ */
$W = [w_i] = [\sum_{j=1}^{q} s_{ij}]$ /* w_i denotes total marks assigned to CLO_i through q question in semester final exam, dimension $n \times 1$ */
$X = [x_i] = [\sum_{j=1}^{q} a_{ij}]$ /* x_i denotes total marks earned to CLO_i through q question in semester final exam, dimension $n \times 1$ */
$R = [r_i] = [\sum_{i=1}^{n} x_i/w_i]$ /* r_i denotes ratio of earned and assigned marks in CLO_i through q question in semester final exam, dimension $n \times 1$ */
$Y = [y_i] = [\sum_{j=1}^{p} c_{ij}]$ /* y_i denotes total marks assigned to CLO_i through p item in continuous assessment, dimension $n \times 1$ */
$Z = [z_i] = [\sum_{j=1}^{p} b_{ij}]$ /* z_i denotes total marks earned to CLO_i through p item in continuous assessment, dimension $n \times 1$ */
$T = [t_i] = [\sum_{i=1}^{n} z_i/Y_i]$ /* t_i denotes ratio of earned and assigned marks in CLO_i through p item in continuous assessment, dimension $n \times 1$ */
Define the linguistic variables (R and T) and their values (Fig. 3)
Develop the membership functions (equation (19), (20) and Fig. 3)
Construct the fuzzy rule base (Table II and III)
Transform input data (R and T) to fuzzy value using the fuzzy membership functions (details in Section III-C1)
Evaluate each of the rules in the rule base (inference, details in Section III-C2)
Combine all the results of those rules (rule aggregation, details in Section III-C2)
Convert the combined results into crisp values called CLO (defuzzification, details in Section III-C3)
$G = [g_{ij}]$ /* g_{ij} is the mapping value between the CLO_i and PLO_j using five levels of values, dimension $n \times m$ */
Generate the PLO using the mapping values and CLOs
Output: CLO, PLO, and Grade

by each of the CLOs in the semester final examination. The second one is the earned marks by each of the CLOs in the continuous assessment. The assigned marks are previously set in the system. Then, the ratios of the earned marks and assigned marks for both are calculated for fuzzification.

This is assuming that there is a total n no. of CLOs for a course of the assessment model. For the semester final, there are q questions to be answered, and for the continuous assessment, p items to be assessed. The marks for each of the questions and items according to the CLO are set by the course teachers and the moderators' panel of the department. Also, the earned marks for each of the students must be traced by the corresponding course teachers. Now, for the semester final, the assigned marks for a student are represented by the matrix S of dimension $n \times q$ shown in (5) as

$$S = [s_{ij}], n \times q \tag{5}$$

where s_{ij} denotes the marks assigned to CLO_i on questions j such that for each of the questions j satisfy the condition in

$$Q = [q_j] = [\textstyle\sum_{i=1}^{n} s_{ij}], 1 \times q. \tag{6}$$

For the continuous assessment assigned marks, another matrix C of dimension $n \times p$ is used as

$$C = [c_{ij}], n \times p \tag{7}$$

where c_{ij} denotes the marks assigned to CLO_i on item j such that for each of the items j satisfy the condition in

$$P = [p_j] = [\textstyle\sum_{i=1}^{n} c_{ij}], 1 \times p. \tag{8}$$

Now, the marks earned by a student on each of the questions and items according to the CLO should be stored by the corresponding course teachers. So, from the semester final exam, get an earned marks matrix A of dimension $n \times q$ represented in (9) as

$$A = [a_{ij}], n \times q \tag{9}$$

where a_{ij} denotes the marks earned to CLO_i on question j.

From the continuous assessment, get another matrix B of dimension $n \times p$ represented in (10) as

$$B = [b_{ij}], n \times p \tag{10}$$

where b_{ij} denotes the marks earned to CLO_i on item j.

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Now, students' performance is evaluated as (5), (7), (9), and (10). For semester final examinations, W^{T} is the summation matrix of assigned marks based on each CLOs of dimension $1 \times n$ generated from

$$W = [w_i] = \left[\sum_{j=1}^{q} s_{ij}\right], n \times 1 \tag{11}$$

where w_i is the total marks assigned on semester final to CLO_i through q questions and W^{T} is the transpose of matrix W. Moreover, w_i can be calculated from the S matrix presented in (5).

And X^{T} is the summation matrix of earned marks based on each CLOs of dimension $1\ x\ n$ from

$$X = [x_i] = \left[\sum_{j=1}^{q} a_{ij}\right], n \times 1 \tag{12}$$

where x_i is the total marks earned on semester final to CLO_i through q questions that can be calculated from (9). Here, X^{T} is the transpose of the matrix X.

Then, using (11) and (12), R^{T} is the ratio matrix to the earned marks and the assigned marks of dimension $1 \times n$ generated from

$$R = [r_i] = \left[\sum_{i=1}^{n} \frac{x_i}{w_i}\right], n \times 1 \tag{13}$$

where $r_i \in [0, 1]$ is the ratio of semester final marks of CLO_i.

Similarly, for continuous assessment, Y^{T} is the summation matrix of assigned marks based on each CLOs of dimension $1 \times n$ generated from

$$Y = [y_i] = \left[\sum_{j=1}^{p} c_{ij}\right], n \times 1 \tag{14}$$

where y_i is the total assigned marks on continuous assessment to CLO_i through p items that can be calculated from (7). Here, Y^{T} is the transpose of the matrix Y.

And Z^{T} is the summation matrix of earned marks based on each CLOs of dimension $1 \times n$ from

$$Z = [z_i] = \left[\sum_{j=1}^{p} b_{ij}\right], n \times 1 \tag{15}$$

where z_i is the total earned marks on continuous assessment to CLO_i through p items that can be calculated from (10). Here, Z^{T} is the transpose of matrix Z.

Then, using (14) and (15), T^{T} is the ratio matrix to the earned marks and the assigned marks of dimension $1 \times n$ generated from

$$T = [t_i] = \left[\sum_{i=1}^{n} \frac{z_i}{y_i}\right], n \times 1 \tag{16}$$

where $t_i \in [0, 1]$ is the ratio of continuous assessment marks of CLO_i. These two ratios, r_i and t_i, are used for fuzzification to evaluate the CLO_i.

1) Fuzzification of Input Variables: There are two input variables (r_i and t_i shown in (13) and (16), respectively) and one output variable (CLO_i) in this evaluation system. The system considers five linguistic values or five levels ($v = 5$) of ratio for the fuzzy sets.

From the fuzzy domain expert and the semester final mark ratios, the system determines the matrix SF of dimension $n \times v$ which indicates the degree of membership of each of the CLOs in the fuzzy domain is symbolized in (17) as

$$SF = [sf_{ij}], n \times v \tag{17}$$

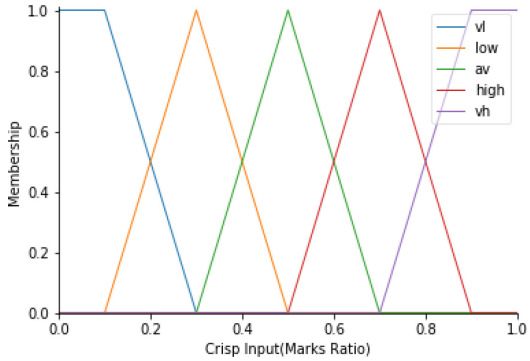


Fig. 3. Membership function for fuzzification: (1) trapezoidal and (2) triangular.

where sf_{ij} denotes the degree of membership for CLO_i and ratio level j in the semester final examination.

Similarly, from continuous assessment ratios, it determines the matrix CA of dimension $n \times v$ indicating the degree of membership of each of the CLOs in the fuzzy domain is symbolized in (18) as

$$CA = [ca_{ij}], n \times v \tag{18}$$

where ca_{ij} denotes the degree of membership for CLO_i and ratio level j in the continuous assessment.

The ratio level $j = 1$ represents the linguistic value "Very Low (vl)," $j = 2$ represents "Low (low)," $j = 3$ represents "Average (av)," $j = 4$ represents "High (high)," and $j = 5$ represents "Very High (vh)."

The membership function is used for the fuzzification shown in (19) and (20), and visualized in Fig. 3. The proposed method uses a mixture of triangular and trapezoidal membership functions as it produces the immediate solution of the optimization problem using a fuzzy system. These functions with the 50% overlap level produce the zero value of the reconstruction error [45]

$$\mu(x, a, b, c, d) = \begin{cases} \frac{x-a}{b-a}, & a \le x \le b \\ 1, & b < x < c \\ \frac{d-x}{d-c}, & c \le x \le d \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

$$\mu(x, a, b, c) = \begin{cases} \frac{x-a}{b-a}, & a \le x \le b \\ \frac{c-x}{c-b}, & b < x < c \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

2) Fuzzy Rules Generation and Aggregation: The proposed fuzzy inference system works based on IF-THEN rules. It has used the Mamdani-type max–min fuzzy inference system. Because it is a well-known and most-used mechanism for inferencing a defuzzification value using the degrees of membership. The system has modeled the rule base for the CLO_i of $R^{\mathrm{T}}(r_i)$ and $T^{\mathrm{T}}(t_i)$ as Table II. All the rules for evaluating the CLO_i are summarized in Table III. The same rules are used for evaluating all the CLOs of the corresponding courses.

Now, for fuzzy rules aggregation, a matrix AG of dimension $n \times v$ is generated as follows:

$$AG = [ag_{ij}], n \times v \tag{21}$$

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TABLE II
FUZZY INFERENCE RULES FOR THE PROPOSED SYSTEM

No.	Rule
1	IF r_i is 'vl' and t_i is 'vl' THEN CLO_i is 'vl'.
2	IF r_i is 'vl' and t_i is 'low' THEN CLO_i is 'vl'.
3	IF r_i is 'vl' and t_i is 'av' THEN CLO_i is 'vl'.
4	IF r_i is 'vl' and t_i is 'high' THEN CLO_i is 'low'.
5	IF r_i is 'vl' and t_i is 'vh' THEN CLO_i is 'av'.
6	IF r_i is 'low' and t_i is 'vl' THEN CLO_i is 'vl'.
7	IF r_i is 'low' and t_i is 'low' THEN CLO_i is 'low'.
8	IF r_i is 'low' and t_i is 'av' THEN CLO_i is 'low'.
9	IF r_i is 'low' and t_i is 'high' THEN CLO_i is 'low'.
10	IF r_i is 'low' and t_i is 'vh' THEN CLO_i is 'av'.
11	IF r_i is 'av' and t_i is 'vl' THEN CLO_i is 'low'.
12	IF r_i is 'av' and t_i is 'low' THEN CLO_i is 'av'.
13	IF r_i is 'av' and t_i is 'av' THEN CLO_i is 'av'.
14	IF r_i is 'av' and t_i is 'high' THEN CLO_i is 'av'.
15	IF r_i is 'av' and t_i is 'vh' THEN CLO_i is 'high'.
16	IF r_i is 'high' and t_i is 'vl' THEN CLO_i is 'av'.
17	IF r_i is 'high' and t_i is 'low' THEN CLO_i is 'av'.
18	IF r_i is 'high' and t_i is 'av' THEN CLO_i is 'high'.
19	IF r_i is 'high' and t_i is 'high' THEN CLO_i is 'high'.
20	IF r_i is 'high' and t_i is 'vh' THEN CLO_i is 'vh'.
21	IF r_i is 'vh' and t_i is 'vl' THEN CLO_i is 'av'.
22	IF r_i is 'vh' and t_i is 'low' THEN CLO_i is 'high'.
23	IF r_i is 'vh' and t_i is 'av' THEN CLO_i is 'vh'.
24	IF r_i is 'vh' and t_i is 'high' THEN CLO_i is 'vh'.
25	IF r_i is 'vh' and t_i is 'vh' THEN CLO_i is 'vh'.

TABLE III
SUMMARY OF FUZZY RULE BASE FOR EVALUATING THE $\mathbf{CLO_i}$

$r_i \backslash t_i$	**vl (1)**	**low (2)**	**av (3)**	**high (4)**	**vh (5)**
vl (1)	vl (1)	vl (1)	vl (1)	low (2)	av (3)
low (2)	vl (1)	low (2)	low (2)	low (2)	av (3)
av (3)	low (2)	av (3)	av (3)	av (3)	high(4)
high(4)	av (3)	av (3)	high(4)	high (4)	vh (5)
vh (5)	av (3)	high (4)	vh (5)	vh (5)	vh (5)

where ag_{ij} is the aggregated value for the CLO_i on its linguistic value j. The value of ag_{ij} can be calculated by (22) as

$$ag_{ij} = \max_{\{(k,l)|\text{Rule}(k,l)=j\}}\{\min(sf_{ik}, ca_{il})\} \tag{22}$$

where *(k,l)*—Rule*(k,l)* $= j$ indicates those rules for which the output CLO_i is j if the input r_i is k and t_i is l from Tables II or III.

3) Defuzzification: The final output values are calculated using a technique called defuzzification. There are so many methodologies to calculate the final crisp output. The proposed system has used the center of area or center of gravity, abbreviated as CoA and CoG method, respectively. So, the CLOs are evaluated using the following equation [46]:

$$CLO_i = CoA = CoG = \frac{\int_{x_{\min}}^{x_{\max}} \mu(x).xdx}{\int_{x_{\min}}^{x_{\max}} \mu(x)dx} \tag{23}$$

where $x_{\min}$ to $x_{\max}$ state the range of linguistic variables, x is the value of that linguistic variables, and $\mu(x)$ is the degree of membership value for a given value of x. So, the CLOs of a course are represented by the matrix CLO of dimension $1 \times n$ denoted in

$$CLO = [df_i], 1 \times n \tag{24}$$

where df_i is the defuzzified crisp value and indicates the CLO_i.

TABLE IV
MAPPING BETWEEN THE CLOS AND THE PLOS: VERY LOW (1), LOW (2), AVERAGE (3), HIGH (4), AND VERY HIGH (5)

CLO\PLO	PLO_1	PLO_2	PLO_3	...	PLO_m
CLO_1	2	1	2	...	4
CLO_2	0	3	5	...	0
CLO_3	3	0	2	...	3
...	...	...	...	...	...
CLO_n	4	1	5	...	5

D. PLO Evaluation Using the CLO

Assume that the n no. of CLOs evaluation are done for a course and the program has m no. of PLOs for evaluation. Then, the PLOs will be evaluated using the values of those CLOs, and the mapping between the CLOs and the PLOs represented in Table IV at different levels. This system considers five levels of mapping for better evaluation purposes. The level values 1, 2, 3, 4, and 5 indicate very lowly, lowly, averagely, highly, and very highly mapped with each other. The zero (0) values indicate a null relation between the corresponding CLOs and PLOs. These mapping values will be set once by the course teachers, and the accreditation committee of the program. Finally, the Academic Committee for Undergraduate Studies (ACUG) will approve it.

The mapped table can be represented by the matrix G of dimension $n \times m$ as in

$$G = [g_{ij}], n \times m \tag{25}$$

where g_{ij} is the mapping level between the CLO_i and the PLO_j.

Now, for the PLO matrix of dimension $1 \times m$ presented in (26) as

$$PLO = [plo_i], 1 \times m \tag{26}$$

where plo_i is the attainment score of PLO_i. This plo_i can be calculated using (27) where the values from (24) and (25) are used

$$plo_i = \begin{cases} \frac{\sum_{j=1}^{n} g_{ji} \times CLO_j}{\sum_{j=1}^{n} g_{ji}}, & \text{if } \sum_{j=1}^{n} g_{ji} \neq 0 \\ 0, & \text{if } \sum_{j=1}^{n} g_{ji} = 0. \end{cases} \tag{27}$$

IV. CASE STUDY

In this section of this article, the background study of the program, known as a case study, will be discussed. It is used to show the evaluation of the CLOs of the courses and PLOs of the program. The Department of CSE at KUET offers two academic programs, namely, Bachelor of Science in CSE (B.Sc. in CSE) and Master of Science in CSE (M.Sc. in CSE).

The B.Sc. in CSE program has a total of 12 PLOs comprising 161 credit hours, four years, and eight semesters. Every year is

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TABLE V
ALL THE COURSE OF EIGHT SEMESTERS FOR EVALUATION

S	Course	Course Title	CLO	S	Course	Course Title	CLO
1st year 1st sem	1. CSE 1101	Structured Programming	07*	3rd year 1st sem	21. CSE 3105	Embedded Systems and IoT	05*
	2. CSE 1107	Discrete Mathematics	07		22. CSE 3107	Applied Statistics...	08
	3. HUM 1107	English and Human Commun...	05*		23. CSE 3109	Database Systems	07*
	4. PHY 1107	Physics	09*		24. CSE 3117	Mobile Computing	08*
	5. MATH 1107	Differential and Integral Cal...	06		25. CSE 3119	Software Engineering...	07*
1st year 2nd sem	6. CHEM 1207	Chemistry	07*	3rd year 2nd sem	26. CSE 3201	Operating Systems	06*
	7. CSE 1203	Digital Logic Design	07*		27. CSE 3209	Artificial Intelligence	06*
	8. CSE 1205	Object Oriented Programming	06*		28. CSE 3211	Compiler Design	06*
	9. EEE 1207	Basic Electrical Engineering	06*		29. ECE 3207	Digital Signal and Image Pro...	07*
	10. MATH 1207	Coordinate Geometry and...	08		30. HUM 3247	Engineers and the Society	05
2nd year 1st sem	11. CSE 2103	Microprocessors and Micro...	07*	4th year 1st sem	31. CSE 4107	Data Communications and...	07*
	12. CSE 2105	Data Structures and Algorithms	07*		32. CSE 4111	Machine Learning	09*
	13. CSE 2113	Computer Architecture	06*		33. CSE 4115	Computer Security	08*
	14. EEE 2117	Analog Electronics	07*		34. CSE 4121	Natural Language Processing	08
	15. MATH 2107	Fourier Analysis and Linear...	09		35. IEM 4127	Industrial Management	09
2nd year 2nd sem	16. CSE 2201	Algorithm Analysis and Design	07*	4th year 2nd sem	36. CSE 4207	Computer Graphics	07*
	17. CSE 2209	Theory of Computation	06		37. CSE 4213	Fault Tolerant System	07
	18. EEE 2213	Digital Electronics	08*		38. CSE 4223	Digital System Design	06*
	19. HUM 2207	Economics and Accounting	06		39. CSE 4233	Robotics	07
	20. MATH 2207	Complex Variable, Vector...	08		40. HUM 4207	Entrepreneurship	06

* S means Semester, the * beside the CLO represents the courses with laboratory.

divided into two semesters. There is a total of 480 students in the department over the four years where 120 students are in each. The department divided the students every year into two sections (60 students per section) for better quality assurance of education. In every semester, five courses are taught, and this research discusses the case study by taking 40 courses as an example shown in Table V. The course code CSE 2113 means CSE for the department, 21 for the second-year first semester, and 01 is the odd number which is for the course. For each of the academic programs, the ABET has a set of criteria that students have to learn throughout the program. The PLOs should have been documented and acknowledged by both the students and the teachers, which will help them to acquire program educational objectives easily. The Board of Accreditation for Engineering and Technical Education (BAETE), Bangladesh, works on accrediting the programs of universities across the country. BAETE has listed about 12 graduate attributes (what graduates have to learn) from the Washington Accord known as PLO for Engineering programs, as shown in Table VII [47]. Generally, they listed the attributes of the engineering programs, but this article has discussed them more specifically for CSE programs. It has also shown a relationship between the PLOs to the Knowledge Profile (K1–K8), the ranges of Complex Engineering Problem Solving (EP1–EP7), and Complex Engineering Activities (EA1–EA5). The complex engineering problem-solving and complex engineering activities are discussed in detail in the BAETE Accreditation Manual for Undergraduate Engineering Programs [48].

This study will focus on evaluating students enrolled in the B.Sc. in CSE program, which includes 40 theoretical and several laboratory courses. First, the case study displays the CLO and PLO assessment of a course as an example and then uses the same procedure for the other courses of a particular student. Finally, the system is run for the assessment of an additional 100 students.

A. Example of CLO and PLO Assessment

Let us take an example for the course CSE 2113 titled Computer Architecture. CSE 2113 has six CLOs as shown in Table VI. There are normally considered six questions to be answered in the semester final examination, each question having 35 marks and 210 marks in total and six items for continuous assessment, such as CT, VV, Quiz, HW, ST, and ATT, where each of the items having 20, 20, 20, 10, 10, and 10 marks, respectively, and 90 marks in total.

Therefore,

No. of CLOs, $n = 6$.

No. of questions in semester final exam, $q = 6$.

No. of items in continuous assessment, $p = 6$.

So, Q, k, P, and h from (1)–(4) are as

$$Q = \begin{bmatrix} q_1 & q_2 & q_3 & q_4 & q_5 & q_6 \end{bmatrix} = \begin{bmatrix} 35 & 35 & 35 & 35 & 35 & 35 \end{bmatrix}$$

$$k = 210,$$

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TABLE VI
CLOs OF THE CSE 2113 COURSE FROM 2ND YEAR 1ST SEMESTER TAKEN AS AN EXAMPLE FOR THE EVALUATION

Course	CLO_i	Descriptions	BLL		
			C	A	P
CSE 2113	CLO_1	Apply the knowledge of performance metrics to find the performance of systems.	3		
	CLO_2	Design a hardware component for an embedded system.	3		4
	CLO_3	Distinguish different types of computer architecture.	2, 3		1
	CLO_4	Measure the performance of different types of computer architecture.	1,5		3, 4, 5
	CLO_5	Analyze the problems in the components of the computer.	4		
	CLO_6	Compare Instruction Level Parallelism and Machine Parallelism.	2, 4, 5	4	

* **BLL (Bloom's Learning Levels):** C: Cognitive, A: Affective, and P: Psychomotor, Numbers indicate the levels of Bloom's skills. For instance, 7 under the P column indicates that the CLO_2 covers the organization level of the psychomotor domain discussed in subsection I-C.

$$\begin{aligned} P &= \begin{bmatrix} p_1 & p_2 & p_3 & p_4 & p_5 & p_6 \end{bmatrix} \\ &= \begin{bmatrix} 20 & 20 & 20 & 10 & 10 & 10 \end{bmatrix} \\ h &= 90. \end{aligned}$$

From (5) and (7), the assigned marks matrices S and C are generated by the course teachers and moderator panel of the department

$$S = [s_{ij}] = \begin{bmatrix} 6 & 12 & 4 & 13 & 8 & 0 \\ 5 & 0 & 9 & 0 & 4 & 13 \\ 7 & 6 & 10 & 6 & 0 & 4 \\ 5 & 7 & 5 & 0 & 11 & 9 \\ 4 & 5 & 7 & 10 & 5 & 3 \\ 8 & 5 & 0 & 6 & 7 & 6 \end{bmatrix}$$

$$C = [c_{ij}] = \begin{bmatrix} 6 & 4 & 4 & 1 & 0 & 0 \\ 5 & 0 & 0 & 0 & 5 & 6 \\ 0 & 8 & 4 & 3 & 3 & 2 \\ 4 & 0 & 3 & 0 & 0 & 1 \\ 5 & 5 & 7 & 2 & 2 & 0 \\ 0 & 3 & 2 & 4 & 0 & 1 \end{bmatrix}.$$

And from (9) and (10), the earned marks matrices A and B are stored by course teachers

$$A = [a_{ij}] = \begin{bmatrix} 3 & 9 & 0 & 12 & 8 & 0 \\ 5 & 0 & 9 & 0 & 3 & 8 \\ 5 & 5 & 6 & 6 & 0 & 4 \\ 3 & 2 & 4 & 0 & 10 & 9 \\ 4 & 5 & 6 & 8 & 5 & 3 \\ 7 & 5 & 0 & 6 & 5 & 6 \end{bmatrix}$$

$$B = [b_{ij}] = \begin{bmatrix} 4 & 2 & 3 & 1 & 0 & 0 \\ 5 & 0 & 0 & 0 & 3 & 5 \\ 0 & 7 & 3 & 2 & 3 & 2 \\ 3 & 0 & 3 & 0 & 0 & 1 \\ 5 & 2 & 5 & 2 & 2 & 0 \\ 0 & 3 & 2 & 2 & 0 & 1 \end{bmatrix}.$$

Now, for the semester final examination, from (11) and (12), the summation matrices W^{T} and X^{T} are calculated as

$$W^{\mathrm{T}} = [w_i] = \begin{bmatrix} 43 & 31 & 33 & 37 & 34 & 32 \end{bmatrix}$$

where $w_1 = 43$ means, the total 43 marks are assigned to CLO_1 through the questions q_1 to q_6

$$X^{\mathrm{T}} = [x_i] = \begin{bmatrix} 32 & 25 & 26 & 28 & 31 & 29 \end{bmatrix}$$

where $x_1 = 32$ means, the total 32 marks are earned from CLO_1 through the questions q_1 to q_6.

Similarly, for the continuous assessment, from (14) and (15), the summations matrices Y^{T} and Z^{T} are calculated as

$$Y^{\mathrm{T}} = [y_i] = \begin{bmatrix} 15 & 16 & 20 & 8 & 21 & 10 \end{bmatrix}$$

where $y_1 = 15$ means, the total 15 marks are assigned to CLO_1 through the items p_1 to p_6.

$$Z^{\mathrm{T}} = [z_i] = \begin{bmatrix} 10 & 13 & 17 & 7 & 16 & 8 \end{bmatrix}$$

where $z_1 = 10$ means, the total ten marks are earned from CLO_1 through the items p_1 to p_6.

Then, from (13) and (16), the ratio matrices R^{T} and T^{T} are computed as

$$\begin{aligned} R^{\mathrm{T}} &= [r_i] \\ &= \begin{bmatrix} 0.744 & 0.806 & 0.788 & 0.757 & 0.912 & 0.906 \end{bmatrix} \quad (28) \\ T^{\mathrm{T}} &= [t_i] \\ &= \begin{bmatrix} 0.667 & 0.812 & 0.850 & 0.875 & 0.762 & 0.800 \end{bmatrix} \quad (29) \end{aligned}$$

here, r_i, $t_i \in [0, 1]$, denote the input or crisp value for the fuzzification generating the CLO_i.

If the crisp values in ratio matrix R^{T} and T^{T} computed in (28) and (29) are given as input to the membership function represented in Fig. 3, and (19) and (20), this will get the degree of membership matrix SF for semester final exam and CA for continuous assessment from (17) and (18) as follows:

$$SF = [sf_{ij}] = \begin{bmatrix} 0 & 0 & 0 & 0.780 & 0.220 \\ 0 & 0 & 0 & 0.470 & 0.530 \\ 0 & 0 & 0 & 0.560 & 0.440 \\ 0 & 0 & 0 & 0.715 & 0.285 \\ 0 & 0 & 0 & 0 & 1.000 \\ 0 & 0 & 0 & 0 & 1.000 \end{bmatrix}$$

here, $sf_{14} = 0.780$ means the degree of membership value for CLO_1 to the ratio level high ($j = 4$) for the semester final exam marks ratio $r_1 = 0.744$ computed in (28)

$$CA = [ca_{ij}] = \begin{bmatrix} 0 & 0 & 0.165 & 0.835 & 0 \\ 0 & 0 & 0 & 0.440 & 0.560 \\ 0 & 0 & 0 & 0.250 & 0.750 \\ 0 & 0 & 0 & 0.125 & 0.875 \\ 0 & 0 & 0 & 0.690 & 0.310 \\ 0 & 0 & 0 & 0.500 & 0.500 \end{bmatrix}$$

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TABLE VII
PLO SET BY BAETE FOR ENGINEERING PROGRAM (DETAILS IN [48])

PLOs	Affiliation	Descriptions
PLO_1	Engineering knowledge	Apply knowledge of mathematics, natural science, engineering fundamentals, and an engineering specialization as specified in K1 to K4 respectively to the solution of complex computer science and engineering problems (EP1 to EP7).
PLO_2	Problem analysis	Identify, formulate, research literature, and analyze complex engineering problems (EP1 to EP7) reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences. (K1 to K4)
PLO_3	Design/development of solutions	Design solutions for complex engineering problems (EP1 to EP7) and design systems, components, or processes that meet specified needs with appropriate consideration for public health, safety, cultural, societal, and environmental considerations. (K5)
PLO_4	Investigation	Conduct investigations of complex computer science and engineering problems (EP1 to EP7) using research-based knowledge (K8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PLO_5	Modern tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex computer science and engineering problems (EP1 to EP7), with an understanding of the limitations. (*K5*)
PLO_6	The engineer and society	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to professional computer science and engineering practice and solutions to complex computer science and engineering problems (EP1 to EP7). (K7)
PLO_7	Environment and sustainability	Understand and evaluate the sustainability and impact of professional computer science and engineering work in the solution of complex computer science and engineering problems (EP1 to EP7) in societal and environmental contexts. (K7)
PLO_8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computer science and engineering practice. (K7)
PLO_9	Individual and teamwork	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PLO_{10}	Communication	Communicate effectively on complex computer science and engineering activities (EA1 to EA5) with the computer science and engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.
PLO_{11}	Project management and finance	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work as a member and leader in a team, to manage projects and in multidisciplinary environments.
PLO_{12}	Life-long learning	Recognize the need for, and have the preparation and ability to engage in, independent and life-long learning in the broadest context of technological change.

* **K1-K8:** Knowledge Profile, **EP1-EP7:** Range of Complex Engineering Problem Solving, **EA1-EA5:** Range of Complex Engineering Activities.

here, $ca_{35} = 0.750$ means the degree of membership value for CLO_3 to the ratio level very high (vh, $j = 5$) for the continuous assessment marks ratio $t_3 = 0.850$ computed in (29).

Now, for obtaining the AG matrix using (22), first, it will consider the CLO_1 in the level of "vl" where $i = 1$ and $j = 1$. Then

$$\begin{aligned} ag_{11} &= \max_{\{(k,l)\mid((1,1),(1,2),(1,3),(2,1))\}}\{\min(sf_{1k}, ca_{1l})\} \\ &= \max\{\min(sf_{11}, ca_{11}), \min(sf_{11}, ca_{12}), \min(sf_{11} \\ &\quad ca_{13}), \min(sf_{12}, ca_{11})\} \\ &= \max\{\min(0, 0), \min(0, 0), \min(0, 0.165), \min(0, 0)\} \\ &= 0. \end{aligned}$$

Similarly, for CLO_1 of other levels, $ag_{12} = 0$, $ag_{13} = 0$, $ag_{14} = 0.780$, and $ag_{15} = 0.220$.

In this way, the system can compute all the values of ag_{ij} for the rest of the five CLOs ($i = 2, 3, 4, 5, 6$), then it gets the AG matrix for defuzzification as

$$AG = [ag_{ij}] = \begin{bmatrix} 0 & 0 & 0 & 0.780 & 0.220 \\ 0 & 0 & 0 & 0.440 & 0.530 \\ 0 & 0 & 0 & 0.250 & 0.560 \\ 0 & 0 & 0 & 0.125 & 0.715 \\ 0 & 0 & 0 & 0 & 0.690 \\ 0 & 0 & 0 & 0 & 0.500 \end{bmatrix}.$$

The defuzzification uses the matrix AG and (23) to evaluate the final CLOs of the CSE 2113 course shown in Fig. 4. Hence, the values in the bold and black line of Fig. 4(a)–(f) crossing at x axes indicate the final output values of CLO_1, CLO_2, CLO_3, CLO_4, CLO_5, and CLO_6, respectively. For

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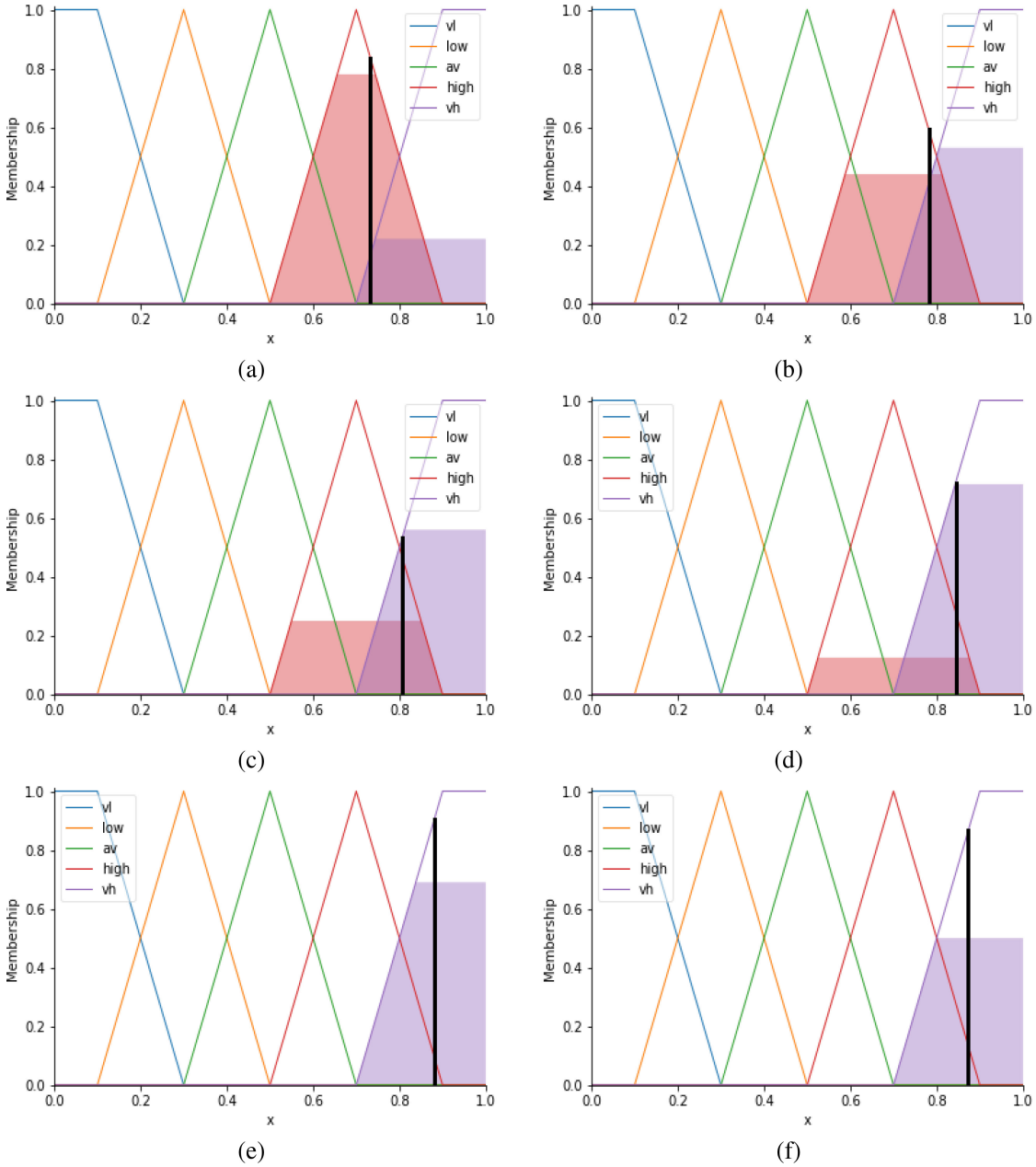


Fig. 4. Defuzzification to evaluate all of the CLOs of CSE 2113 – Computer Architecture course, line crossing the x-axis indicates the attainment, (a) attainment of CLO_1, (b) attainment of CLO_2, (c) attainment of CLO_3, (d) attainment of CLO_4, (e) attainment of CLO_5, and (f) attainment of CLO_6.

evaluating the CLO_1, (23) integrates the colored areas of the "high" and "vh" in Fig. 4(a), where the y-axis values of them are 0.780 and 0.220, respectively, used from the AG matrix. In the same way, other CLOs, especially CLO_5 [Fig. 4(e)] and CLO_6 [Fig. 4(f)], are evaluated using the values of 0.690 and 0.500, respectively, from the AG matrix. The whole procedure is implemented using the SciKit-Fuzzy package in Python [49]. So, the CLO matrix is as follows:

$$CLO = \begin{bmatrix} 0.733 & 0.782 & 0.807 & 0.844 & 0.881 & 0.873 \end{bmatrix}.$$

The research considers the system for the CSE program, and the program has $m = 12$ PLOs shown in Table VII. The course CSE 2113 has $n = 6$ CLOs. So, from Table VIII, G represented in (25) will be as

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{30}$$

Now, considering $i = 1$, the attainment of PLO_1 is calculated as follows using (27):

$$plo_1 = \frac{\begin{array}{c} g_{11} \times CLO_1 + g_{21} \times CLO_2 + g_{31} \times CLO_3 \\ + g_{41} \times CLO_4 + g_{51} \times CLO_5 + g_{61} \times CLO_6 \end{array}}{g_{11} + g_{21} + g_{31} + g_{41} + g_1 + g_{11}}$$

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TABLE VIII
CLO TO PLO MAPPING FOR PLO EVALUATION (THIS IS THE SAME FOR ALL THE STUDENTS FOR A PARTICULAR COURSE)

Code	CLO_i	PLO_1	PLO_2	PLO_3	PLO_4	PLO_5	PLO_6	PLO_7	PLO_8	PLO_9	PLO_{10}	PLO_{11}	PLO_{12}
CSE 2113	CLO_1	1	0	0	0	0	0	0	0	0	0	0	0
	CLO_2	0	0	4	0	0	0	0	0	0	0	0	0
	CLO_3	0	0	0	3	0	0	0	0	0	0	0	0
	CLO_4	0	2	0	0	0	0	0	0	0	0	0	0
	CLO_5	0	4	0	0	0	0	0	0	0	0	0	0
	CLO_6	0	0	0	2	0	0	0	0	0	0	0	0

TABLE IX
ABSOLUTE GRADING SYSTEM (OUT OF 4.00)

Marks%	**Letter**	**Grade**	**GP(out of 4.00)**
80% or above	A^+	A plus	4.00
75% to less than 80%	A	A	3.75
70% to less than 75%	A^-	A minus	3.50
65% to less than 70%	B^+	B plus	3.25
60% to less than 65%	B	B	3.00
55% to less than 60%	B^-	B minus	2.75
50% to less than 55%	C^+	C plus	2.50
45% to less than 50%	C	C	2.25
40% to less than 45%	D	D	2.00
less than 40%	F	Fail	0.00

* GP: Grade Point.

$$plo_1 = \frac{1\times 0.733 + 0\times 0.782 + 0\times 0.807 + 0\times 0.844 + 0\times 0.881 + 0\times 0.873}{1+0+0+0+0+0} = 0.733.$$

Similarly, the other PLOs are evaluated, and the matrix PLO is acquired

$$PLO = \begin{bmatrix} 0.733 & 0.869 & 0.782 & 0.833 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Now, using the steps described in the methodology section and the above example, all of the CLOs for each of the courses are evaluated. As the evaluation system has taken 40 courses as an example shown in Table V for the case study, the CLO evaluation of those courses is presented in Table X. It can be noticed that in Table V, the maximum number of CLOs for a single course is 9. That is why Table X shows the evaluation of CLO_1 to CLO_9. The average CLOs are also calculated to compare the attainment of corresponding courses with the existing method. The mapping between CLOs and PLOs of a course from 2nd year 1st semester is shown in Table VIII. Then, using (27) and Tables VIII–X, the PLOs attainment in each course are calculated, which is displayed in Table XI. However, Table VIII contains the mapping of only one course, and others exist in the dataset. A total of 12 PLOs are listed in Table VII for engineering education, so 12 PLOs are evaluated. The highlighted attainments mark the student's learning on which CLOs/PLOs he/she did exceptionally well (Min 80% attainment). This defines the student's expertness in corresponding CLOs/PLOs, which can help in their professional life. The PLOs are also averaged to compare the changes in the other semester and for the final attainment at the end of the program.

V. RESULT ANALYSIS

The main objective of this system is to provide an evaluation approach for the OBE model through attaining the CLOs and PLOs of courses by the students throughout their graduation period. This model helps the engineering program to examine the advantages and limitations to evaluate a comprehensive educational process that assists students in maximizing opportunities for the attainment of learning outcomes. The OBLT is necessary for both teachers and students. The teacher can find out their shortcomings at the time of teaching in the classroom. On the other hand, students can measure whether they have achieved their target level or not. For implementing the whole methodology, the Mamdani-type fuzzy-logic-based method is used. Initially, 40 courses have been considered to evaluate a student. Later, the system is run on an additional 100 students. In OBE, the students have an opportunity to develop themselves continuously. First, let us introduce the traditional grading methods practiced in most Bangladeshi universities, especially at KUET. This percentage-based grading system is summarized in Table IX. The course teachers and moderators set questions based on their syllabus without considering Bloom's taxonomy, learning methods, CLOs, PLOs, ABET standards, etc. The course teachers simply store the total earned marks by the student and do not need to trace the marks based on CLOs. The percentage of the total earned marks from the total assigned marks in a corresponding course is calculated by (31). From the proposed system, this can be computed from the ratio of the earned marks matrix shown in (5) and (7), and the assigned marks matrix shown in (9) and (10).

Now, the method evaluates the percentages of marks using (31) for all of the 40 courses that have been taken for case study purposes. The grades of traditional evaluation are generated in Table XII

$$\text{Obtained Marks\%} = \frac{\text{Total earned marks}}{\text{Total assigned marks}} \times 100\% = \frac{\sum A + \sum B}{\sum S + \sum C} \times 100\%. \quad (31)$$

This can be seen from Table XII that for the courses CSE 1101 (1) and CSE 3105 (21), the student gets 194 and 224

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TABLE X
DEMONSTRATION OF CLO ATTAINMENT FOR ALL THE COURSES OF A PARTICULAR STUDENT'S PERFORMANCE EVALUATION THROUGHOUT THE EIGHT SEMESTERS

Sem.	*Course*\CLO_i	CLO_1	CLO_2	CLO_3	CLO_4	CLO_5	CLO_6	CLO_7	CLO_8	CLO_9	*Average*
1st year 1st sem	1. CSE 1101	0.644	0.728	0.722	0.699	0.605	0.595	0.633	–	–	0.661
	2. CSE 1107	0.878	0.779	0.796	0.775	0.779	0.807	0.781	–	–	0.799
	3. HUM 1107	0.757	0.857	0.832	0.833	0.79	–	–	–	–	0.814
	4. MATH 1107	0.728	0.682	0.718	0.624	0.67	0.448	–	–	–	0.645
	5. PHY 1107	0.789	0.711	0.774	0.815	0.793	0.837	0.681	0.676	0.724	0.756
1st year 2nd sem	6. CHEM 1207	0.587	0.715	0.632	0.705	0.667	0.83	0.671	–	–	0.687
	7. CSE 1203	0.583	0.724	0.646	0.706	0.605	0.637	0.667	–	–	0.653
	8. CSE 1205	0.733	0.782	0.807	0.844	0.881	0.873	–	–	–	0.82
	9. EEE 1207	0.733	0.558	0.693	0.573	0.796	0.715	–	–	–	0.678
	10. MATH 1207	0.779	0.802	0.882	0.865	0.834	0.878	0.7	0.888	–	0.828
2nd year 1st sem	11. CSE 2103	0.608	0.679	0.663	0.666	0.538	0.654	0.644	–	–	0.636
	12. CSE 2105	0.583	0.724	0.646	0.706	0.605	0.637	0.667	–	–	0.653
	13. CSE 2113	0.733	0.782	0.807	0.844	0.881	0.873	–	–	–	0.82
	14. EEE 2117	0.6	0.675	0.674	0.656	0.719	0.536	0.743	–	–	0.658
	15. MATH 2107	0.805	0.881	0.892	0.794	0.774	0.862	0.681	0.778	0.745	0.801
2nd year 2nd sem	16. CSE 2201	0.647	0.608	0.622	0.706	0.591	0.743	0.624	–	–	0.649
	17. CSE 2209	0.666	0.765	0.769	0.757	0.733	0.747	–	–	–	0.74
	18. EEE 2213	0.748	0.757	0.888	0.875	0.867	0.85	0.737	0.872	–	0.824
	19. HUM 2207	0.741	0.877	0.815	0.832	0.779	0.765	–	–	–	0.801
	20. MATH 2207	0.742	0.781	0.86	0.889	0.774	0.853	0.652	0.757	–	0.788
3rd year 1st sem	21. CSE 3105	0.709	0.778	0.783	0.75	0.798	–	–	–	–	0.764
	22. CSE 3107	0.679	0.797	0.842	0.848	0.774	0.878	0.64	0.842	–	0.787
	23. CSE 3109	0.696	0.642	0.641	0.656	0.522	0.698	0.618	–	–	0.639
	24. CSE 3117	0.757	0.883	0.887	0.876	0.876	0.858	0.737	0.892	–	0.846
	25. CSE 3119	0.621	0.642	0.605	0.774	0.585	0.631	0.725	–	–	0.655
3rd year 2nd sem	26. CSE 3201	0.686	0.885	0.793	0.878	0.742	0.826	–	–	–	0.802
	27. CSE 3209	0.748	0.798	0.737	0.753	0.711	0.6	–	–	–	0.724
	28. CSE 3211	0.769	0.69	0.782	0.634	0.664	0.622	–	–	–	0.694
	29. ECE 3207	0.604	0.668	0.669	0.822	0.684	0.549	0.61	–	–	0.658
	30. HUM 3247	0.733	0.757	0.813	0.785	0.811	–	–	–	–	0.78
4th year 1st sem	31. CSE 4107	0.6	0.737	0.685	0.737	0.575	0.575	0.805	–	–	0.673
	32. CSE 4111	0.853	0.886	0.883	0.81	0.867	0.833	0.777	0.796	0.774	0.831
	33. CSE 4115	0.79	0.794	0.878	0.869	0.873	0.875	0.56	0.888	–	0.816
	34. CSE 4121	0.761	0.802	0.882	0.865	0.834	0.878	0.7	0.88	–	0.825
	35. IEM 4127	0.841	0.816	0.892	0.886	0.515	0.84	0.774	0.774	0.886	0.803
4th year 2nd sem	36. CSE 4207	0.531	0.774	0.83	0.524	0.456	0.731	0.656	–	–	0.643
	37. CSE 4213	0.346	0.393	0.427	0.354	0.548	0.404	0.351	–	–	0.403
	38. CSE 4223	0.654	0.806	0.857	0.818	0.774	0.849	–	–	–	0.793
	39. CSE 4233	0.6	0.79	0.628	0.882	0.651	0.584	0.737	–	–	0.696
	40. HUM 4207	0.825	0.787	0.818	0.89	0.886	0.874	–	–	–	0.847

* The highlighted attainments indicate the student's learning on which CLOs he/she did exceptionally well (Min 80% attainment). This defines the student's expertness in corresponding CLOs, which can help in his professional life.

marks, respectively. Their percentages of marks are 64.67% and 74.7% which are very close to 65% and 75%. As the existing evaluation is followed by an absolute grading system, the student gets grades B and A minus, respectively, which is not fair and satisfactory for the student. It may be a matter of a sorrowful, distressing situation for the students. Because if he/she got one or two more marks in these two courses, respectively, their grades would have been upgraded by one grade. So, the questions in the absolute grading system, the course teachers may be biased toward the student or may go in favor or may show some kindness to the students in that case, and may add one or a few extra marks to that student. However, in the proposed CLO-based evaluation system using fuzzy logic, the teachers do not know the grades

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TABLE XI
DEMONSTRATION OF PLO ATTAINMENT FOR ALL THE COURSES OF A PARTICULAR STUDENT'S PERFORMANCE EVALUATION THROUGHOUT THE PROGRAM

Course\PLO_i	PLO_1	PLO_2	PLO_3	PLO_4	PLO_5	PLO_6	PLO_7	PLO_8	PLO_9	PLO_{10}	PLO_{11}	PLO_{12}
1. CSE 1101	0.728	0.722	0.693	0	0.660	0	0	0	0	0	0	0.610
2. CSE 1107	0.804	0.791	0.781	0	0	0	0	0	0	0	0	0
3. HUM 1107	0	0	0	0	0.824	0	0	0	0	0.814	0	0
4. MATH 1107	0.601	0.586	0	0	0	0	0	0	0	0	0	0
5. PHY 1107	0.748	0.748	0	0	0	0	0	0	0	0	0	0
6. CHEM 1207	0.67	0.715	0	0	0	0	0.671	0	0	0	0	0
7. CSE 1203	0.596	0.641	0.655	0	0.637	0	0	0	0.655	0	0	0
8. CSE 1205	0.807	0.835	0.853	0	0.82	0	0	0	0.877	0	0	0.749
9. EEE 1207	0.71	0.644	0	0	0	0	0	0	0	0	0	0
10. MATH 1207	0.814	0.827	0	0	0	0	0	0	0	0	0	0
11. CSE 2103	0.608	0.625	0.657	0.538	0.655	0	0	0	0	0	0	0
12. CSE 2105	0	0.624	0.652	0.667	0.667	0	0	0	0	0	0	0.619
13. CSE 2113	0.733	0.869	0.782	0.833	0	0	0	0	0	0	0	0
14. EEE 2117	0.67	0.592	0.708	0	0.743	0	0	0	0	0	0	0
15. MATH 2107	0.778	0.837	0	0	0	0	0	0	0	0	0	0
16. CSE 2201	0.628	0.636	0.652	0.668	0.591	0	0	0	0	0	0	0
17. CSE 2209	0.733	0.711	0.757	0	0	0	0	0	0	0	0	0
18. EEE 2213	0.753	0.876	0.813	0	0.857	0	0	0	0	0	0	0
19. HUM 2207	0.75	0.822	0	0	0	0	0	0	0	0	0.782	0
20. CSE 2207	0.735	0.818	0	0	0	0	0	0	0	0	0	0
21. CSE 3105	0.739	0.791	0.791	0.798	0.798	0.791	0	0.783	0	0	0	0
22. CSE 3107	0.793	0.824	0.828	0.848	0.774	0	0	0	0	0	0	0
23. CSE 3109	0.649	0.648	0.621	0.658	0.656	0	0	0	0	0	0	0
24. CSE 3117	0.836	0.848	0.861	0.804	0.879	0	0.892	0	0	0	0	0
25. CSE 3119	0.621	0.711	0.681	0.662	0.616	0	0	0	0	0	0	0
26. CSE 3201	0.705	0.862	0.882	0.826	0.742	0	0	0	0	0	0	0
27. CSE 3209	0.734	0.74	0.763	0.644	0.753	0	0	0	0	0	0	0
28. CSE 3211	0.769	0.634	0.666	0.723	0.69	0	0	0	0	0	0	0
29. ECE 3207	0.65	0.644	0.684	0.549	0.662	0	0	0	0	0	0	0
30. HUM 3247	0	0	0	0	0	0.745	0.813	0.785	0	0.811	0	0
31. CSE 4107	0.643	0.683	0.575	0.64	0.575	0	0	0	0	0	0	0
32. CSE 4111	0.819	0.841	0.884	0.822	0.837	0	0	0	0	0	0	0
33. CSE 4115	0.859	0.843	0.88	0.724	0.881	0	0	0	0	0	0	0
34. CSE 4121	0.813	0.822	0.842	0.88	0.831	0	0	0	0	0	0	0
35. IEM 4127	0.778	0.874	0	0	0.515	0	0	0	0	0	0.886	0
36. CSE 4207	0.658	0.634	0.656	0.687	0.596	0	0	0	0	0	0	0
37. CSE 4213	0.348	0.434	0.404	0.464	0	0	0	0	0	0	0	0
38. CSE 4223	0.736	0.794	0.815	0.796	0.818	0	0	0	0	0	0	0
39. CSE 4233	0.594	0.67	0.779	0.882	0.618	0	0	0	0	0	0	0
40. HUM 4207	0.8	0.815	0	0.89	0.874	0.874	0	0	0	0.874	0.866	0
Average	0.714	0.738	0.736	0.727	0.725	0.810	0.790	0.785	0.766	0.833	0.845	0.659

* The highlighted attainments mark the student's learning on which PLOs he/she did exceptionally well (Min 80% attainment). This defines the student's expertness in corresponding PLOs, which can help in his professional life.

of the students. Hence, no bias and kindness cases occur in the course teacher for the student. On the other hand, from the comparison between absolute grading and the proposed evaluation shown in Table XII, it can be seen that the problem described for absolute grading is solved. In the courses, CSE 2105 and CSE 3105, the average CLO evaluations in percentage are upgraded to 65.3% and 76.4%, respectively. If it converts these CLO evaluations into grades, then the students will get B plus and A, which were B and B minus in the absolute grading system, respectively. So, the proposed method

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TABLE XII
FUZZY-LOGIC-BASED CLO EVALUATION AND THE EXISTING EVALUATION FOR A PARTICULAR STUDENT

Serial Index	Course Code	Earned Marks	Marks / 300	Average CLO	Existing Grade	Proposed Grade
1	CSE 1101	194	0.647	0.653	B	B plus
2	CSE 1107	240	0.800	0.799	A plus	A
3	HUM 1107	243	0.810	0.814	A plus	A plus
4	MATH 1107	193	0.643	0.645	B	B
5	PHY 1107	223	0.743	0.756	A minus	A
6	CHEM 1207	202	0.673	0.687	B plus	B plus
7	CSE 1203	194	0.647	0.653	B	B plus
8	CSE 1205	242	0.807	0.82	A plus	A plus
9	EEE 1207	203	0.677	0.678	B	B
10	MATH 1207	240	0.800	0.828	A plus	A plus
11	CSE 2103	192	0.64	0.636	B	B
12	CSE 2105	194	0.647	0.653	B	B plus
13	CSE 2113	242	0.807	0.820	A plus	A plus
14	EEE 2117	195	0.652	0.658	B plus	B plus
15	MATH 2107	234	0.78	0.801	A	A plus
16	CSE 2201	195	0.65	0.649	B plus	B
17	CSE 2209	220	0.733	0.74	A minus	A minus
18	EEE 2213	234	0.78	0.824	A	A plus
19	HUM 2207	235	0.783	0.801	A	A plus
20	MATH 2207	229	0.763	0.788	A	A
21	CSE 3105	224	0.747	0.764	A minus	A
22	CSE 3107	224	0.747	0.787	A minus	A
23	CSE 3109	195	0.65	0.639	B plus	B
24	CSE 3117	240	0.8	0.846	A plus	A plus
25	CSE 3119	195	0.65	0.655	B plus	B plus
26	CSE 3201	235	0.783	0.802	A	A plus
27	CSE 3209	210	0.7	0.724	A minus	A minus
28	CSE 3211	210	0.7	0.694	A minus	B plus
29	ECE 3207	193	0.643	0.658	B	B plus
30	HUM 3247	231	0.77	0.78	A	A
31	CSE 4107	200	0.667	0.673	B plus	B plus
32	CSE 4111	236	0.787	0.831	A	A plus
33	CSE 4115	239	0.797	0.816	A	A plus
34	CSE 4121	240	0.800	0.825	A plus	A plus
35	IEM 4127	232	0.753	0.803	A	A plus
36	CSE 4207	191	0.635	0.643	B	B
37	CSE 4213	118	0.393	0.403	F	D
38	CSE 4223	240	0.800	0.793	A plus	A
39	CSE 4233	203	0.677	0.696	B plus	B plus
40	HUM 4207	242	0.807	0.847	A plus	A plus

is an unbiased, fair student evaluation technique that provides reliability and robustness. Similarly, the courses in the blue color also get increased to the next upper grades. The CSE 4213 (37) course failed by gaining 39.3% marks and passed through the proposed system.

The reverse cases may also occur in the system. From Table XII, this is also noticed that for the courses CSE 1107 (2), CSE 2201 (16), CSE 3109 (23), CSE 3211 (28), and CSE 4223 (38) in red marked get A plus, B plus, B plus, A minus, and A plus, respectively, in the existing grading system. But in the proposed evaluation system, the grades become A, B, B, B plus, and A, which are decreased by one grade. It is one of the demerits of the system if the average CLOs of the courses are converted to the corresponding grades. So, the CLO and PLO-based OBE is required where a student's performance will be evaluated by the attainment of the CLOs of a course and PLOs of the program. Few more cases also occur in this system, such as in the CSE 3209 (27) and CSE 3211 (28), the earned marks are 210 for each course but CLO attainments are 0.724 (grade A), and 0.694 (grade B plus), respectively. This is due to the unequal earned marks at several CLOs, which is important to note, as different CLOs have a different level of importance to learning.

The statistics from Table XII show that the CLO attainment of that student increased by 34 courses from percentage-based mark attainment. On the other hand, the attainment decreased by only six courses. From those statistics, this is also to be noted that among the 34 courses, there is a total of 15 grades that have been upgraded to the next upper grade. And among the six courses, there is a total of five grades that have been degraded to a previous lower grades. Though this grade upgradation or degradation is not important in OBE, this is illustrated for the system performance evaluation. In addition, OBE is not assessed based on the grade but based on CLO–PLO attainment.

The system has also resulted in the PLOs of the courses for evaluating students' attainments. The CLOs and PLOs are mapped by five levels of importance described in the previous section. For the ABET accreditation purposes department of CSE, KUET proposed the CLO to PLO mapping where the mapping level is binary [either fully(1) or not(0)]. In this binary level of mapping, the CLO and the PLO attainment will be the same if these are fully mapped, though they do not provide any methodology to demonstrate the CLOs and PLOs. However, this binary mapping should not be used to evaluate. So, this system has proposed five levels of mapping for better evaluations. The average PLO attainment is evaluated as in Fig. 5. Table XI is prepared from Table X and the mapping data between the CLOs and PLOs of a course. The mapping for CSE 2113 (13th course) is shown in Table VIII, where it can be noticed that this course covers only four PLOs (PLO_1–PLO_4). That is why Table XI contains only four PLOs evaluated for the CSE 2113 course. Similarly, the CSE 4213 (37th course) is evaluated with low-scored CLOs in Table X, so the PLOs are low-scored in Table XI. On the other hand, the proposed system is a general and robust framework that can be easily evaluated by changing the mapping data in Table VIII, to extend the evaluation frequency of PLO_6–PLO_{12}.

A. Statistical Analysis

In the previous sections, the case study has been done in 40 courses for a particular student. This system is also run on

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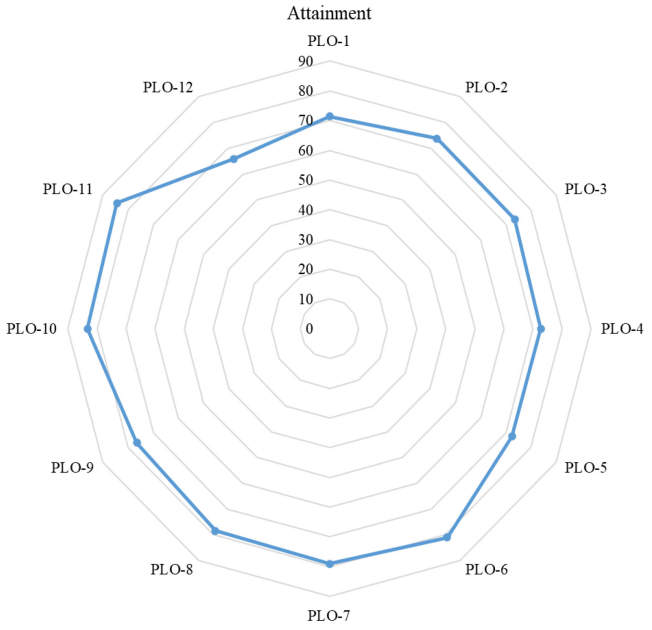


Fig. 5. Average PLO attainment for the courses of all eight semesters for a particular student, which should be learned by the student during the period of graduation.

100 students' data, evaluating their performance for statistical analyses. The CLO attainment level for each of the 40 courses for each of the 100 students is generated. Table XIII shows the statistical changes of those 100 students of four-year courses. The statistics are demonstrated in terms of CLO attainment changes and corresponding grade changes compared with the percentage-based earned marks. From Table XIII, it is noticed that for students 1, 2, and 3, the CLO attainment increased in 22, 24, and 22 courses, while it decreased in 18, 16, and 18 courses, respectively. The no. of grade changes for these students is upgraded by 2, 5, and 6 courses and is degraded by 2, 2, and 2, respectively. Overall, this is better for the students as the total number of grade upgrades is greater than the degradation. This has used the usual rules from Table III where the weight of the ratio in the semester final examination is greater than continuous assessment. The system also generated the PLO attainment level for each of the 40 courses using the CLO attainment for 100 students. The average of the PLO attainment is calculated indicating the final PLO attainment that the students should learn after the end of the graduation period.

TABLE XIII
STATISTICS OF CLO ATTAINMENT LEVEL AND GRADE FOR 100 STUDENTS

SI	Attainment change		Grade change	
	increased	decreased	upgraded	degraded
Student-1	22	18	2	2
Student-2	24	16	5	2
Student-3	22	18	6	2
Student-4	23	17	5	2
Student-5	20	20	5	6
Student-6	20	20	2	3
Student-7	27	13	3	2
Student-8	25	15	6	3
Student-9	22	18	4	3
Student-10	18	22	2	3
Student-11	26	14	5	2
Student-12	27	13	5	1
Student-13	20	20	4	2
Student-14	23	17	3	1
Student-15	19	21	4	5
Student-16	17	23	2	4
Student-17	21	19	3	2
Student-18	21	19	4	1
Student-19	28	12	4	1
Student-20	20	20	5	3
Student-21	27	13	6	2
Student-22	19	21	3	1
Student-23	16	24	1	5
Student-24	25	15	4	2
Student-25	24	16	5	2
...	...	...	...	...
Student-50	22	18	2	4
Student-51	18	22	6	4
Student-52	25	15	7	2
Student-53	20	20	4	3
Student-54	17	23	5	4
Student-55	27	13	9	1
Student-56	25	15	5	2
Student-57	17	23	0	3
Student-58	20	20	2	1
Student-59	26	14	3	2
Student-60	25	15	3	4
Student-61	19	21	2	5
Student-62	20	20	7	2
Student-63	22	18	3	2
Student-64	24	16	4	1
Student-65	27	13	4	3
Student-66	28	12	7	2
Student-67	26	14	2	1
...	...	...	...	...
Student-100	17	23	4	4

VI. DISCUSSION

The course learning and PLO-based student evaluation are internationally accepted standards for getting accreditation, especially for engineering students. Moreover, to improve the academic program and student quality, a large number of organizations or programs (departments) of an organization are applying for ABET accreditation. The most important challenge here is to face the problem of scarcity of well-organized documentation, implementation of computing programs for the successful accreditation process, methodology of academic performance evaluation, etc. Considering this lack of resources, there are needed a sufficient number of employees to design documented methodologies and collect educational organograms practiced by different institutions on their steps

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toward the accreditation process. In the context of ABET criteria, one of the most important facets is that institutions must have a course learning or program learning OBE. The department of CSE of KUET is Currently working on accreditation processes. They have prepared documentation considering ABET criteria, but a well-documented student evaluation process. The absolute grading system for evaluation is kept as they practice it but are trying to move forward for CLO–PLO evaluation.

Regarding a proper student evaluation, this article has provided an outcome-based evaluation technique for computing the CLOs and PLOs of a course and program, respectively. The method is implemented to compute the CLOs and PLOs by the fuzzy-logic system on the way toward ABET accreditation. As most of the universities in Bangladesh consider continuous assessments consisting of the CT, VV, quiz, STs, HW, class ATTs, assignments, etc., and the semester final examination for evaluation, the proposed method also considers these two parameters for the evaluation. The continuous assessment evaluation is done throughout the entire semester, and the semester final exam is taken after the end of the semester. From the methodology analyses, it can be noticed that the weight (marks) of the semester final exam is much higher than the continuous assessment. This is considered during the fuzzy rule generation. First, the CLOs of a course are calculated from the earned marks and the assigned marks for the students using the fuzzy inference system. Second, from the CLOs and the mapping weight of CLO to PLO, it has generated the PLOs and the corresponding grade.

A. Why Does the Method Use Fuzzy Inference System?

Recently, FIS has been widely used not only for educational purposes but also for practical and commercial purposes, such as in the medical [50], industrial [51], etc., sectors. Fuzzy always gives acceptable reasoning, other than precise reasoning. FIS and intelligent systems both are coherent with each other as FIS consists of rule bases and membership functions by which it can decide something on many engineering problems. Moreover, fuzzy logic deals with uncertainty, especially in the engineering sector. In the education sector of the traditional hard-boundary-based deterministic approach, an evaluator may easily know the grade of a student as they know the earned marks of the student. So, If the student gets short of a few marks for a better grade, the evaluator may get biased for the student. As fuzzy logic works on uncertainty there is no way to know the grade of the students. FIS is a robust system as it works on several types of data, including imprecise, vague, and distorted data. Additionally, there is no need for a huge amount of data to implement the fuzzy algorithms, so they do not require extra memory space. On the other hand, it is more flexible to add and delete fuzzy rule bases in FIS based on problem-solving requirements. That is why the system uses fuzzy logic for the outcome-based student performance evaluation.

B. Challenges Have to Meet

Approaching such an evaluation system, it is most important to have available, adequate, experienced, and trained faculty members associated with the entire procedure, such as preparing questions considering Bloom's taxonomy and CLO, storing marks to evaluate, etc. So, for a developing country like Bangladesh, it is most challenging to provide experienced faculty members, where most of the faculty members are lecturers or assistant professors level. The supporting staff is also inadequate in most of the universities here. Another issue is that faculty members at Higher Education Institutions (HEIs) frequently come from diverse academic backgrounds, and the proposed system's complexity may make understanding the fuzzy set difficult. The goal is to present a comprehensive and adaptable framework that can be used in a range of academic applications so as not to complicate the evaluation process.

VII. Conclusion

This research proposes a student performance evaluation technique. This evaluation technique will help the students to track, measure, and plan for their continuous quality improvement. This can also assist the institutional quality monitoring cell in tracking and evaluating the key terms of OBE, and planning for continuous quality improvement. In addition, the accreditation bodies of an institution can use this evaluation method as a guideline for the homogeneous and coherent assessment and certification of all the programs over the universities. For practically implementing the evaluation technique, there have to be some extra challenges. Such as the departments have to ensure the course CLOs are being set following Bloom's taxonomy, learning methodology, the language of learning outcomes, etc., and question papers are being prepared by considering those parameters. The main challenge here is the need for more effort from the course teachers to prepare the earned marks of each student and store them. Though it is a drawback of the system, it will help to yield more reliable, meaningful, and fair grades for the students. Moreover, this OBE-based evaluation technique greatly helps to get recognition and accreditation of the program all over the world.

Future research in this field may go in different directions that are closely aligned to the OBE process and evaluation. This study shows the evaluation of the students considering his/her marks and mapping between the CLOs and PLOs on an Excel sheet. The university can design a desktop application or Web-based software for the automation of the entire system. The OBE components must be integrated with the system, such as the CLOs of each course, PLOs of the program, their mapping, and assigned and earned marks based on CLOs. The FIS must also be combined with the software/application. This automated evaluation system will clear the way for the CLOs and PLOs attainment. Therefore, the academic transcript of OBE certification can easily be generated automatically from the system. Additionally, some other important parameters can be derived from this system for quality measurement. The students may identify their progress reports graphically for each course at the end of each semester. The system can also summarize a student's learning on which CLOs and PLOs he/she did exceptionally well. This tracking of learning defines the student's expertise in diverse engineering activities and tools for real-life problem solving and determines the

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moral and ethical point of view while serving the country. The software/application should support both the OBE-based evaluation and the existing evaluation based on predefined criteria. Future researchers can also add some modules for faculty members' evaluation to develop both learning and teaching skills.

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CONFLICTS OF INTEREST

The authors declare that there are no conflicts of interest related to this publication.

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Abdul Aziz received the bachelor's and master's degrees in computer science and engineering from the Khulna University of Engineering and Technology (KUET), Khulna, Bangladesh, in 2017 and 2022, respectively.

He worked as a Lecturer with the Department of Computer Science and Engineering, KUET, from January 2018 to December 2020, where he is currently working as an Assistant Professor. He is also working on the OBE curriculum of the Department of Computer Science and Engineering, KUET. His research interests include artificial intelligence, deep learning, machine learning, computer vision, fuzzy logic, and engineering education.

Md. Asaf-uddowla Golap received the B.Sc. and M.Sc. Engineering degrees in computer science and engineering from the Khulna University of Engineering and Technology (KUET), Khulna, Bangladesh, in 2017 and 2019, respectively.

From December 2020 to May 2022, he was a Lecturer with the Institute of Information and Communication Technology, KUET, where he is currently working as an Assistant Professor. His current research interests include artificial intelligence, deep learning, machine learning, genetic programming, biomedical signal processing, and fault tolerance.

M. M. A. Hashem received the Ph.D. degree in artificial intelligence systems from Saga University, Saga, Japan, in 1999.

He is currently a Professor with the Department of Computer Science and Engineering, Khulna University of Engineering and Technology (KUET), Khulna, Bangladesh. He is also actively involved in developing an outcome-based education curriculum for the Department of Computer Science and Engineering, KUET. He has authored over 100 peer-reviewed publications in international journals and conferences. His research interests include artificial intelligence, machine learning, IoT security, health informatics, evolutionary computation, and soft computing.

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